Employee Attrition Analysis – Executive Summary

Overview

This project analyzed employee attrition data to identify the key factors that influence whether employees stay with or leave the company. The goal was to provide actionable insights that can help HR teams improve employee retention, reduce turnover costs, and maintain workforce stability.

Key Findings

The analysis revealed that employee attrition is influenced by a combination of compensation, workload, job satisfaction, and career-related factors. Employees who frequently work overtime, have lower job satisfaction, and receive lower monthly income were significantly more likely to leave the organization.

The Sales department experienced the highest employee turnover rate among all departments. Within job roles, Sales Representatives showed the highest likelihood of leaving, making them a priority group for retention efforts. The data also indicated that employees in their first few years with the company are at a higher risk of attrition compared to long-tenured employees.

While salary plays an important role in employee retention, it is not the only factor driving attrition. Employees with poor work-life balance, limited career growth opportunities, and lower job satisfaction were also more likely to resign, even when compensation was competitive.

Predictive Model Results

Three machine learning models were developed and compared to predict employee attrition. The best-performing model successfully identified employees at higher risk of leaving and highlighted the factors most strongly associated with turnover. These predictions can help HR teams take proactive measures before employees decide to leave.

Recommendations

1. Focus retention efforts on high-risk departments and roles, particularly Sales Representatives and employees who frequently work overtime.

2. Improve employee engagement by conducting regular satisfaction surveys, offering career development opportunities, and providing clear growth paths within the organization.

3. Strengthen work-life balance initiatives through flexible scheduling, workload management, and employee wellness programs.

4. Develop targeted retention programs for employees during their first five years with the company, as this group shows the highest attrition risk.

Conclusion

Employee attrition is driven by multiple factors rather than salary alone. By focusing on employee satisfaction, workload management, career development, and early retention strategies, the company can significantly reduce turnover and improve overall workforce stability. The predictive model developed in this project can serve as a valuable decision-support tool for HR teams when identifying employees who may require additional engagement and retention efforts.